An AI-Powered Platform to Bridge the Gap Between Sri Lankan Citizens and Government Services GOV Connect Lk

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specializing in Information System Engineering

Department of Information Technology
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DECLARATION

I hereby declare that the research proposal is my original work and has been prepared by me. The information and data presented in this report are true and accurate to the best of my knowledge. All sources of information and literature used in this report have been properly cited and acknowledged.

I further declare that this proposal has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution, and it complies with the ethical standards of research as outlined by Sri Lanka Institute of Information Technology.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of supervisor:- Date :- 27/08/25

Signature of co-supervisor:- Date :- 28/08/2025

ABSTRACT

Complaint redressal in Sri Lanka is still slow, manual and inefficient, causing delays, misclassification and lack of prioritization of issues. Citizens often lodge grievances in informal Sinhala-English code-mixed text, that further complicates automated systems and excludes rural communities with low digital literacy. To fill these gaps, this project is proposing GovConnect LK, an Artificial Intelligence (AI) powered complaint management tool that will use Natural Language Processing (NLP), Machine Learning (ML) to make government service delivery more efficient, transparent and responsive. The neural network language (NLP) engine trained on Sinhala-English and code-mixed datasets automatically categorizes the complaints and also identifies urgency levels. A complementary ML model is trained to predict expected resolution time given past or simulated data. These outputs, along with community voting signals, drive a dynamic priority score to more intelligently handle complaints. Complaints and classifications are centralised in a database and are shown via a real-time officer dashboard, empowering government officials to easily monitor, prioritise, and resolve grievances. The innovation in the research is how it can take as input the informal multilingual data, how it has a hybrid prioritization algorithm for the data, and how it brings citizen feedback to the decision-making. In conclusion, GovConnect LK has the potential to bring faster, fairer, and more inclusive public grievance management to Sri Lanka, setting an example of AI-driven governance in developing regions.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
GPU	Graphics Processing Unit
HCI	Human-Computer Interaction
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
API	Application Programming Interface
UI	User Interface
UX	User Experience
WCAG	Web Content Accessibility Guidelines

1. INTRODUCTION

1.1. Background of the Research

In Sri Lanka, citizens' voices are often not heard within the prevailing mechanisms of public complaints management. For decades, the process has been tied to paper forms, telephone calls, and manual Government officer routes, which is slow, inconsistent and prone to misclassification. While the name grievance management suggests a cordial intention to solve issues of the public in a time-efficient manner, in reality such a system has not been achieved. One of the most immediate problems is language. Citizens, especially in rural and semi-urban areas, often post complaints in a code-mixed way in Sinhala and English, which is a natural reflection of people's speech. But existing systems are not smart enough to understand such multilingual and colloquial expressions. As a result, frequent complaints are misconstrued, misdirected, or entirely ignored, leaving serious issues unaddressed and eroding the public's confidence in governmental bodies. This gap highlights a broader issue of Sri Lanka's reliance on outdated, manual systems which not only leads to slow response times, but also limits access to low digital-literacy communities. People's voices - voices of the marginalized that need efficient governance the most - are swallowed in the system's inefficiencies. It is to this end that this research puts forth GovConnect LK, a futuristic solution to redefine grievance management using the capabilities of artificial intelligence. Using Natural Language Processing (NLP), GovConnect LK is designed to categorize complaints in Sinhala, English or code-mixed, to make sure each voice is correctly decoded and directed to the right channel. A part of this work involves the research and development of a fine-tuned Sinhala-English NLP model that can be sensitive to informal and low-resource language inputs to enable real-time classification of citizen complaints. More than a technological product, GovConnect LK signifies a leap towards inclusive governance, where citizens are empowered to express themselves in a natural way and governments are equipped to listen efficiently. In this way, the project aims at implementing a complaint management system that is swifter, fairer and transparent, which will build the basis for enhanced citizen participation and a restoration of confidence in the public service delivery system.

1.2. Literature Survey

Although the study of complaint management in the realm of public services has evolved considerably, empirical literature on the actual application of these procedures in low-resource and code-mixed settings, such as Sri Lanka, is scarce. Some research finds that structured complaint systems have a measurable impact on increased trust and efficiency of a Malaysian Public Complaints Bureau; however, there is no indication that research has embedded natural language processing (NLP) based classification at the point of intake [1]. Similarly, evaluation studies in Malaysia using the DMAIC framework identified accessibility and responsiveness as important satisfaction drivers but fell short in automating intake for citizens text that is code-mixed or informal [2][3]. Many recently launched AI-powered grievance platforms offer pipelines for automatic classification and urgency detection, with proven advantages in terms of scale and responsiveness. As such, the existing case studies such as India's Rail MADAD have shown that NLP categorization and emergency detection can speed up routing and enhance service in a monolingual and domain-specific setting, rather than multilingual Sri Lankan complaints [4][5][6][7]. Experience in governance contexts (e.g. Jakarta) has shown that effective complaint systems improve transparency and open-government outcomes, but few incorporate AI or multilingual intake in the public-facing layer [8][9][10]. Sinhala-English code mixing is a serious problem. A large body of literature outlines code-mixed data to be limited in low-resource availability, ambiguous tokens, borrowing and intra-word mixing which complicate classification tasks [14][15][16]. While studies on token-level Language Identification (LID) show that models such as CRFs and sequence taggers perform better than generic classifiers, few studies apply these techniques to downstream complaint classification in real time. Finally, some studies [17] have experimented with normalization and neural machine translation (NMT) for

Sinhala-English code-mixed texts, verifying that pre-processing and curated corpora is necessary, but it remains data dependent and limited. Hybrid and transformer-based models have proven to be promising. Recent studies on multilingual transformers like XLM-R and Sinhala-BERT have shown promising results in sentiment analysis and code-mixed classification with the provision of interpretable rationales using explainability (e.g., SHAP, LIME) [18][19]. However, these studies are largely experimental in nature and so far fail to operate explainable urgency detection within government complaint systems. Similarly, public-sector NLP reviews describe the adoption process as having three stages of automation, extension, and transformation, with classification and triage identified as "automation level wins" (although few examples are given in Sri Lanka's multilingual context [11][12]). Under continuous improvement models, learning optimization is identified as a problem for public service management. Quality management literature shows that DMAIC-type loops can help improve involvement and effectiveness when used in tandem with analytics, yet there is a dearth of multilingual NLP-based complaint triage integration [3]. Healthcare complaint studies also identify expectation gaps related to improper triage and especially slow response times, showing early and systematic categorisation is not only technical but also essential for keeping citizen trust [13]. These gaps show a need for Sinhala-English code-mixed NLP solutions, for Sri Lankan users.

1.3. Research Gap

The research gap is that there is a need to adapt the traditional complaint management and classification systems for the multilingual and code-mixed environment in Sri Lanka, especially dealing with complaints made in Sinhala-English code-mixed text (Singlish). This can be interpreted in the sense of the following points.

Re A: This study, which was carried out based on the evaluation of complaints at the Malaysian Public Complaints Bureau, showed that structured complaint pipelines increase service trust and efficiency [1]. Although it does address accessibility and responsiveness, it does not address the problems of code-mixed or informal digital text input, restricting its usefulness to the linguistic environment of Sri Lanka.

Research B: NLP based classification and urgency detection in AI-based grievance platforms such as Rail MADAD in India [6][7] These were efficient for routing but limited to monolingual and domain-specific complaint categories and did not support low-resource Sinhala-English code-mixed data or explainability for trust in government workflows.

Research C: Sentiment analysis and code-mixed classification tasks have been adapted from multilingual NLP research including transformer-based models [18][19], e.g., XLM-R and Sinhala-BERT. While encouraging, this approach is experimental and there is no existing domain-aligned taxonomies for Sri Lankan ministries, nor interpretable urgency detection which is critical for citizen trust and administrative decision making.

Such limitations discussed would mean that a system that embeds Sinhala-English code-mixed handling, urgency-aware triage and explainability within complaint classification would lend itself to provide a more inclusive and transparent public-service platform. This would provide cultural and linguistic understanding, facilitate real-time intake, as well as bolster trust amongst citizens through interpretable outputs.

Feature	**Research 1**	**Research 2**	**Research 3**	**Proposed**
Handles Sinhala–English Code-Mixing	✗	✗	Partial	✓
Domain-Aligned Complaint Taxonomies	✗	Partial	✗	✓
Urgency Detection for Routing	✗	✓	Partial	✓
Explainable AI for Trust (SHAP/LIME)	✗	✗	Partial	✓
Policy & Governance Alignment (Sri Lanka)	✗	✗	✗	✓

Table 1: Research Gap

1.4. Research Problem

AI-powered grievance management systems are a significant step over manual and semi-automated manual systems a decade ago as they support smart complaint classification monitoring in real-time aligned with service delivery contexts. Such systems have, however, been implemented only in a handful of use cases worldwide and their adaptation to multilingual, code-mixed complaints in Sri Lanka has largely not been addressed. This lack of localization constrains the delivery of inclusive, efficient, and trust-based citizen services, especially as Sri Lankan citizens often communicate in Sinhala-English code mixed ("Singlish"), often informal and non-standard text.
Complaint monitoring with automated classification and urgency-aware triage in real time has the following advantages:

It provides basic efficiency gains in complaint handling, particularly in terms of being able to highlight urgent or high-risk issues that are not apparent in traditional manual sorting [1].

It helps to understand nuanced urgency cues in short and code-mixed text that are hard to capture by generic NLP models without special approach to preprocessing and training [14][15].

It enables justifiable reasoning via explainable AI methods (e.g. SHAP/LIME), which is important in government workflows for officer trust and accountability [18].

It gives further information to implement the "service gaps" of traditional systems, ensuring that the less privileged citizens with lower digital literacy will not be excluded due to language barriers [2][3].

It is an addition to governance processes that is driven by the evidence-based decision-making and connects automated NLP intake to continuous service improvement structures like DMAIC [11].

One of the main challenges is that Sinhala-English code-mixed complaints are token-level ambiguous, borrowing and irregularly spelled so commonly misclassified by general NLP datasets and multilingual models when used directly. This leads to misguided routing decisions and degradation of effectiveness. For instance, an informal Singlish complaint describing a critical health or safety issue could be classified which would delay critical response interventions. Additionally, existing systems rarely use domain-specific taxonomies mapped to Sri Lankan ministries and authorities that are necessary for proper routing and accountability. This shortcoming is exacerbated by the lack of culturally annotated training corpora, with most current models trained on generic or foreign datasets that do not accurately depict the characteristics of Sri Lankan citizens' discourse [16][17].

The strictness of Sri Lanka's public-service system adds to the difficulty: often, complaints cut across institutional lines and need to be coordinated and specific. Citizens often experience delays and misclassification, which undermines trust and leads to use of informal channels. Cognitive trust in government services is known to be enhanced when outputs from artificial intelligence (AI) are explainable and transparent, but most existing natural-language processing (NLP)-based complaint systems do not feature such explainability. Even understanding of service trends and citizen engagement is limited by the lack of longitudinal dashboards to visualize a breakdown of complaint categories, urgency levels, and response pattern over time [8][9].

To summarize, there is a lack of a dedicated system for the classification of complaints in multiple languages with code-mixing, with the ability to detect the urgency and explainability of the complaints, which is one of the main gaps in the public complaint management of Sri Lanka. Overcoming this will enable citizens to engage in government more meaningfully without the kind of hindrances that are currently in place, and will accelerate the development of a more responsive, transparent and inclusive complaints management ecosystem in Sri Lanka.

2. OBJECTIVES

2.1. Main Objective

This research aims at creating an NLP based system that automatically classifies Sinhala, English and code-mixed citizen complaints and guarantees faster, consistent, and efficient handling of grievances to improve the government responsiveness.

2.2. Sub Objectives

The five important sub objectives are given below:

- To clean, normalize, and address spelling variations, slang, and colloquial expressions in a dataset of citizen complaints in Sinhala, English, and code-mixed Sinhala-English.
- Further, we fine-tune multilingual transformer models (i.e., mBERT, XLM-R), to categorize multilingual code-mixed complaints in Sinhala-English for accurate classification.
- To deploy the trained NLP model into actual-time complaint management platform with a user-friendly interface available to the government officers and citizens alike.
- To measure the system against accuracy, precision, recall and F1-score metrics and benchmark against existing methods of manual complaint classification.

To deploy the system in an e-governance environment with the capability to deliver efficient grievance redressal, prompt citizen interaction and greater transparency in complaint resolution.

3. METHODOLOGY

The study has a systematic approach to advance an NLP-based complaint classification system that can read Sinhala-English code-mixed citizen complaints in real-time. Phase 1 is data collection, with the collection of real-world complaints from government portals, social media platforms, emails and surveys. These complaints include Sinhala, English and code-mixed text.

Each complaint is manually classified into pre-defined categories, such as infrastructure, healthcare and public services, to generate a supervised learning dataset. Once collected, the data is pre-processed to enhance quality and model readiness. This includes cleansing the text, that is removing irrelevant characters and formatting flaws, normalization of spelling variants, and dealing with slang or colloquial terms. Tokenization is done to break text into words or subwords with special care taken to preserve both Sinhala and English tokens in code-mixed text. In the model development phase, appropriate multilingual transformers (mBERT or XLM-R) models are chosen and then fine-tuned for the classification of informal, code-mixed complaints. Hyperparameters such as learning rate, batch size, and number of epochs are tuned to maximize the model's accuracy and reliability.

In the system design and integration phase, a citizen portal is created in which citizens can submit complaints via the Internet. Next, this NLP model is integrated to classify these complaints automatically in real-time. Also, there is a dashboard for government officials to review and handle the classified complaints in an efficient manner.

The system is rigorously evaluated and validated using performance metrics such as accuracy, precision, recall, and F1-score to assess the model's performance. Comparative analysis is conducted against existing manual classification methodologies and the feedback from officers and citizens is gathered to determine usability and effectiveness. Finally, deployment and testing in the wild are included in the research.

The system is tested in selected government offices or online portals, and the performance is monitored with respect to classification accuracy, complaint processing time and user satisfaction. The NLP model and the system interface are continually refined and enhanced to ensure practical, inclusive and efficient use of the solution for real-world e-governance applications in Sri Lanka by using available feedback and performance data.

3.1. System Architecture Diagram

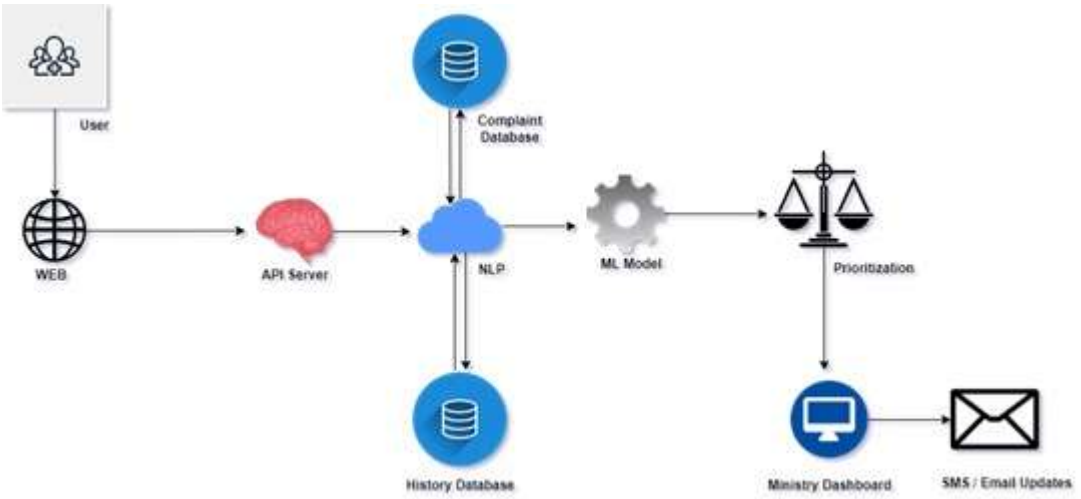

Figure 1: System Architecture Diagram

The digital system for handling complaints, GOV Connect LK, was built to handle citizen issues in urban and rural Sri Lanka. The platform supports English and code-mixed languages, to give access to all the citizens, residents to report on issues like hospital staff shortages, potholes, water supply, missing garbage collection, and issues with school infrastructure. It also links with Ministries such as health, education, highways, water supply, environment, public administration, transport, power and energy, agriculture and labor.

GOV Connect LK consists of two elements. First, a language processing engine that automatically analyzes the text of the complaint, without the need for an additional software application, and automatically categorizes it into categories (e.g., health, roads, water) and urgency (high, medium, low). The flagged complaints can thus be sent through to the relevant ministry. Second, a data-driven model which identifies past complaints and forecasts completion times and feasibility factors to determine how long the complaint could take and how feasible it is to do so again, helping to prioritize complaints that will have the most impact.

The mechanism also allows people to participate in the process by voting on the complaint posted by others and providing greater visibility to issues that many others are experiencing (they might even define the issue differently from each other). All submitted complaints are ranked by the ministry dashboards by urgency, time of expected resolution and number of votes.

This process allows transparency, faster resolution of problems, and better delivery of public services to the community and will give citizens a larger voice concerning government.

3.2. Component specific System Diagram

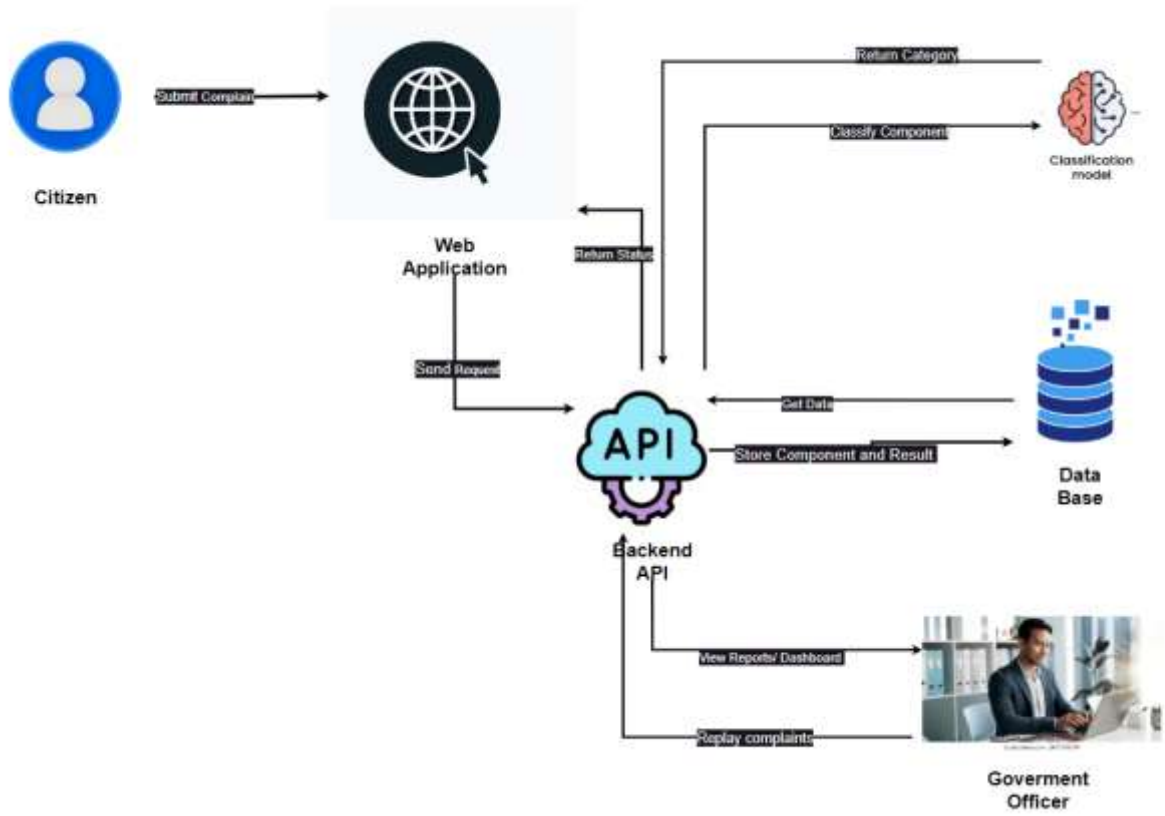

Figure 2: Component Specific System Diagram

The component wise system diagram shows the end-to-end process of managing citizen's complaints in an intelligent and systematic way. The system starts with the Citizen module where citizens submit their grievance in text. These submissions are then sent to the Text Preprocessing component which ensures that the raw data of complaint is cleaned, normalized, and tokenized to be ready for correct analysis. The processed text is sent to the Complaint Classification module that uses computational techniques to classify the complaint into predefined types or classes that make the complaint more understandable and actionable. Once the classification of the complaint is done, the system performs several concurrent actions: Through the Notification module, when the alert is generated automatically, the concerned officer will get the notification and will transfer the matter to the concerned authority without any delay. In tandem, the structured complaint is sent to the Dashboard module, giving a clear and organized view of all complaints in real-time, helping administrators and officers to make decisions and keep track of all of the complaints. At the same time, the Data Management component will ensure that the classified complaint is placed into the database, to create a source of reliable information for future references, analysis, and reporting. This integrated flow ensures that the complaint of each and every citizen is not just captured and processed efficiently but also classified, stored, visualized and directed to the right officer at the right time. By linking submission, pre-processing, classification, storage, notification and visualization, the system facilitates a seamless and transparent grievance redressal process that enhances accountability, responsiveness, and service delivery.

3.3. Project Requirements

3.3.1. Functional Requirements

Function name	Input	Processing	Output
Complaint Submission	Complaint text (Sinhala, English, or code-mixed), citizen details	Accept complaints via web/mobile interface, validate and store submission	Confirmation message with unique complaint ID
Text Preprocessing	Raw complaint text	Clean, normalize, tokenize text; handle spelling, slang, and colloquial expressions	Preprocessed text ready for classification
Complaint Classification	Preprocessed complaint text	Apply NLP model, categorize complaint into predefined categories (e.g., infrastructure, health)	Classified complaint with category label
Dashboard for Officers	Classified complaints database	Provide structured view, enable search, filtering, and prioritization	Interactive dashboard with categorized complaints
Real-Time Notifications	New complaint submission	Send alert to relevant officers; notify citizen of successful submission and classification	Notification to officers and citizens
Reporting and Analytics	Complaint records and system logs	Generate summaries, track frequency, resolution status, and model performance metrics	Reports and analytics dashboards
User Authentication & Roles	Login credentials (for officers/admins)	Verify credentials, assign role-based permissions	Secured access for officers/admins; tracking for citizens with complaint ID
Data Management	Complaint records, classification results	Store securely, allow CRUD operations by admins	Updated and managed complaint database

Table 2: Functional Requirement

3.3.2. Non-Functional Requirements

NFR	Testable NFR
Performance: The system should process and classify complaints in real-time with minimal delay.	Each complaint shall be classified in under 2 seconds. The system shall support at least 100 concurrent submissions without noticeable performance degradation.
Scalability: The system should accommodate future growth in users, complaints, and categories.	The system shall handle a 50% yearly increase in complaints without requiring architectural changes, and support addition of new languages/categories with <10% extra latency.
Usability: The interface must be intuitive for citizens and efficient for officers.	Citizens with minimal technical skills shall be able to submit a complaint within 3 steps, and officers shall filter/search complaints in the dashboard within 2 clicks.
Reliability & Availability: The system should remain continuously operational with minimal downtime.	The system shall maintain 99.5% uptime annually and ensure no more than 5 minutes of downtime per month. Backup and recovery shall ensure zero data loss in failures.
Security: Sensitive complaint data must be protected.	All complaint records shall be encrypted (AES-256 at rest, TLS in transit). Officers must access the system through role-based authentication, with failed login attempts locked after 3 tries.
Maintainability: The system should allow easy updates and bug fixes.	All modules shall be documented with API references, enabling updates or model retraining within 2 hours of deployment downtime. Bug fixes shall be patchable within one release cycle.
Portability: The platform should work across devices and operating systems.	The complaint portal shall be accessible on desktop, tablet, and smartphone, and compatible with Chrome, Firefox, Edge, and Safari with at least 95% identical functionality.
Accuracy: The NLP model must achieve reliable classification.	The complaint classification accuracy shall be at least 85% F1-score for Sinhala-English code-mixed text, with misclassification not exceeding 10% of total complaints.

Table 3: Non-Functional Requirement

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3.4. Tools and Technologies

3.4.1. Tools

1. Programming Environment

JavaScript and Python are the main programming languages. For the frontend and backend development, JavaScript is used, and Python is used for training NLP models and for preprocessing the text. Visual Studio Code (VS Code) is an integrated development environment, and it offers debugging, syntax highlighting, and extensions for both JavaScript and Python development.

2. Frontend

The frontend was created with React.js and Vite is the build tool. React: This tech empowers the development of dynamic and responsive UIs for complaint submission and officer dashboards. Vite prioritizes fast development, hot module replacement, and efficient bundling, accelerating overall development speed and user experience. CSS is used for styling with standard CSS or CSS frameworks like Tailwind CSS.

3. Backend

Node.js with Express.js serves as a backend server, managing API requests, routing, and interfacing with the front end and database. For real-time classification of the complaints, REST API is used to integrate the NLP model with the UI.

4. Database

MongoDB is the database used to store the complaint records, user data and model outputs. Its document-oriented structure is flexible and allows unstructured and semi-structured data storage, which is critical to the management of code-mixed Sinhala-English complaints.

5. Machine Learning and NLP Model development

is done in Python using Hugging Face Transformers for fine-tuning of pre-trained multilingual models like mBERT, XLM-R, PyTorch and TensorFlow for model training and evaluation. NLTK and spaCy are used for text pre-processing, tokenization, and normalization.

6. Data Handling and Analysis

Pandas and Numpy are applied to clean and manipulate the datasets, data transformation and feature extraction for the machine learning pipeline.

7. Version Control and Collaboration

Git and GitHub is used for version control, collaborative programming and maintaining multiple branches of the project. 8. Visualisation and Reporting Model results are plotted with matplotlib and seaborn, and dashboards and reports on complaint patterns and system performance are created in Power BI.

3.4.2. Technologies

Frontend

In essence, the frontend layer is built using React.js and Vite as a build tool. This combination of technology enables the construction of dynamic and interactive user interfaces for citizen complaint submissions and complaint management by government officials. React's component-based architecture allows for modular and reusable components, while Vite allows for faster development and live feedback during deployment. React and Vite will facilitate a responsive and effective interface with the citizen and government officer.

Backend

The backend layer is implemented using Node.js with Express.js. Through the use of javascript and Express.js, the back end supports the scalable, server-side logic and API routing necessary to process complaints into the system. Using Node.js and Express.js, the system can handle multiple incoming requests simultaneously and effectively manage the communication between the back end, the front end, and a database. Making use of Express.js makes it easy to create RESTful APIs which enables simple integration with the NLP model at the time of making a complaint and provides immediate complaint classification.

Database

The database technology used for the project is MongoDB, which will store structured and unstructured complaint data. As a document-oriented database, MongoDB can easily store complaint records that may contain code-mixed Sinhala-English text and has a flexible format to store complaint records. As a NoSQL file-based database, it is inherently scalable, has faster querying of document data which is advantageous when managing many complaints submitted by citizens.

NLP and Machine Learning

Discussion: The research applies Python-based NLP frameworks (Flair/NLTK/Transformers/pandas) and implement the complaint classification system. Since it deals with multilingual and code-mixed Sinhala-English complaints, fine-tuning of multilingual transformer models (mBERT and XLM-R) is used to generate embedded vector representation for accurate complaint classification in multiple languages, including code-mixed context. Deep learning frameworks (e.g., TensorFlow and PyTorch) are applied for training and fine-tuning and evaluation of the models used in the complaint classification system. The scope of model performance needed for real world datasets warrants more than simply relying on machine learning frameworks. Specifically, the algorithms being used in this system, described as follows, are critical for achieving accurate classification in a general sense, but also as part of the context multilingual, code-mixed complaint classification:

•Transformer models (BERT, mBERT, XLM-R) with embedding layers are used for contextual embedding and multilingual text classification tasks in a way that identifies semantic relationships that cross the boundaries of multiple languages.

•Fine-tuning algorithms are important for this study as they use existing pre-trained models of transformed models on the complaint dataset, to make them even more relevant to the complaint dataset, while becoming

more accurate.

•Tokenization and normalization algorithms are important for spelling variation within the complaint as well as the normalizing and processing of the stopwords, and slang or colloquial talk into the final complaint classification.

•Vectorization/embedding algorithms encode the original textual input to a representational numerical input for interpreting by a neural network frame.

•Classification algorithms. Each prediction of the complaint categories comes from softmax layers inside the transformers for classifying complaints.

•Evaluation metrics algorithms: Evaluate performance by using accuracy, precision, recall and F1-score to ensure reliability and effectiveness of the system.
Together all of the above-mentioned technologies and algorithms can provide a strong NLP framework that is able to process informal, code-mixed Sinhala-English complaints for real-time classification and e-governance applications.

Visualisation and Reporting

Visualisation technologies are used to understand the model results and generate insights. Data plotting frameworks and reporting tools provide capabilities for reporting to allow you to track complaint patterns, model accuracy and classification trends. This allows stakeholders to evaluate the system performance to support decision making for e-governance applications.

3.5. Implementation

The result of the proposed solution is a web-based interactive platform for classifying code-mixed Sinhala-English complaints in real-time. The system combines frontend visualization, backend processing, NLP based classification and database management for accurate and effective handling of complaints for citizens and government officers.

Frontend Development

The frontend is built using Vite and React.js, which provides a responsive and interactive experience for the dashboard where complaints can be submitted, tracked, and monitored. The modular nature of React.js enables the interface to be implemented in components including complaint forms, categorized view and status dashboards. Virtual DOM and state management enables the application to update instantly, when a new complaint is submitted the data is immediately updated in the dashboard. Tailwind CSS is used for uniform and easy styling. The frontend also provides accessibility features, to make it usable across devices for all users.

Backend Development

The backend is built using Node.js and Express.js as an API layer between frontend, NLP model and database. Express.js is used to process complaint submission, retrieval and classification requests in real-time. Highly parallelizable backend architecture to handle multiple simultaneous requests for efficient communication between system components Integration with the NLP model will ensure classification of complaints immediately upon submission to aid the government officers in making timely decisions.

NLP Model Implementation

The NLP component is based on Python-based mBERT and XLM-R multilingual transformer models that have been fine-tuned using code-mixed Sinhala-English complaints datasets. Preprocessing methods such as tokenization, normalization, lemmatization, and feature extraction are used to prepare the data for model training. Model training, optimization, and evaluation are done using deep learning frameworks such as TensorFlow and PyTorch. Complaints are defined into well-defined categories by our NLP model, which supports ongoing monitoring and eases the efficiency of e-governance applications.

Database Management

MongoDB is used for holding structures as well as unstructured complaint data. Its document-based structure provides flexibility in code mixed text handling with fast query/retrieval/update capabilities. The database is tightly integrated with the backend, which allows complaint records, user details, and the outputs from the NLP model to be stored securely. With the cloud hosting solution, there is support for several concurrent users and real-time access to data for citizens as well as government officers.

3.6. Deployment

We deploy the complaint classification system to ensure the seamless integration between the AI-based complaint classification model and the web application to deliver real-time analysis of code-mixed Sinhala-English complaints. The platform will house trained transformer models as well as an interactive frontend-backend-based interface that will allow citizens and government officers to visualize and engage with the data and information provided by the model.

Deployment will include the following:

Model Hosting: The optimized mBERT and XLM-R transformer models will be hosted on Hugging Face Spaces while utilizing the GPU for efficient real-time classification of complaints. This configuration makes it possible to have low latency inference where the backend handles incoming complaints directly without the need to compute locally on the user side. Hugging Face also permits easy integration of the APIs with the backend application developed using Node.js and Express.js.

Web Hosting: The frontend and the backend will be containerized using Docker along with the React.js frontend and Node.js backend hosted on a cloud platform. This provides uniformity in functionality across all environments and secure and reliable access to users. The system will be integrated with MongoDB to efficiently handle complaint data and provide seamless delivery of dashboards, complaint submission form and real-time classification results.

Scalability & Access: The system supports 100+ concurrent users with cloud-based scaling, keeping performance high during peak loads. It is available on desktop and mobile browsers and features role-based access control so that officers are given administrative privileges and citizens can submit and track their complaints. The design is focused on usability and accessibility to deliver an inclusive experience for all users across both urban and rural areas.

Thus, the deployment provides a secure, scalable, and accessible platform that can not only classify complaints in real-time but dynamically scale for future growth as part of the e-governance ecosystem in Sri Lanka.

3.7. Evaluation of the system

Evaluation of the complaint classification system is conducted to test the technical performance and usability, with the objective of ascertaining the accurate, reliable and practical implementation of the NLP-based platform for classification of code-mixed Sinhala-English complaints in real-time. The evaluation is done by using a mix of quantitative indicators (accuracy, precision, recall, F1 score and classification performance) and qualitative feedback from end-users (citizens and government officers).

Accuracy, precision and recall performance of the system is obtained on the basis of classification performance (accuracy, precision, recall). Precision calculates the percentage of complaints correctly classified to a category out of all the predicted complaints for that category. Recall is the percentage of complaints correctly classified divided by all actual number of complaints in that category. Target accuracy and recall is expected to be high (>85%) to guarantee the reliability of classification especially, considering the code-mixed Sinhala-English text with informal and colloquial expressions.

- F1 Score

The F1 score, which is a weighted average of precision and recall, is a single metric that captures the model's performance in classifying complaints. With a target F1 score of >85%, the model provides a balance between precision and coverage, confidently providing predictions for real-world mixed-language complaints.

• Real-Time Performance and Responsiveness.

The system is validated to process the incoming complaints in real-time. The assessment guarantees that the NLP model can classify complaints without a significant delay, facilitating quick decision-making for the government officers. Metrics include average time to respond to a complaint and throughput under concurrent requests.

• Intervention Effectiveness (usability of the system)

The system is evaluated by interactions with government officers and simulated citizen reporting of complaints to demonstrate the practical usability of the system. This assessment looks at operational efficiency of complaint handling and navigation and classification accuracy. The goal is to improve the efficiency of processing complaints by a minimum of 80%, the system helps to reduce manual labour and enables consistent classification.

• User Satisfaction

Qualitative feedback is gathered from citizens and government officers by means of surveys and usability tests. This assessment is based on system accessibility, ease of understanding of dashboards, ease of submitting a complaint, and overall satisfaction with real-time classification. The goal is to have satisfaction levels of over 80% and make the platform usable, user-friendly and usable by different user groups.

Consequently, the evaluation is based on the combination of quantitative performance metrics and qualitative user feedback to ensure that the complaint classification system is technically robust and user-centric. High accuracy, precision, recall, F1 score and user satisfaction prove the reliability, efficiency and usability of the system for its real-life application in e-governance applications.

4. WORK BREAKDOWN STRUCTURE

The Work Breakdown Structure (WBS) for Complaint Classification System research is presented below. It breaks down the project into manageable components such as initial preparation, requirement gathering, documentation, implementation, system integration, and validation. This ensures that all critical tasks are planned, resources are allocated efficiently, and progress is monitored systematically.

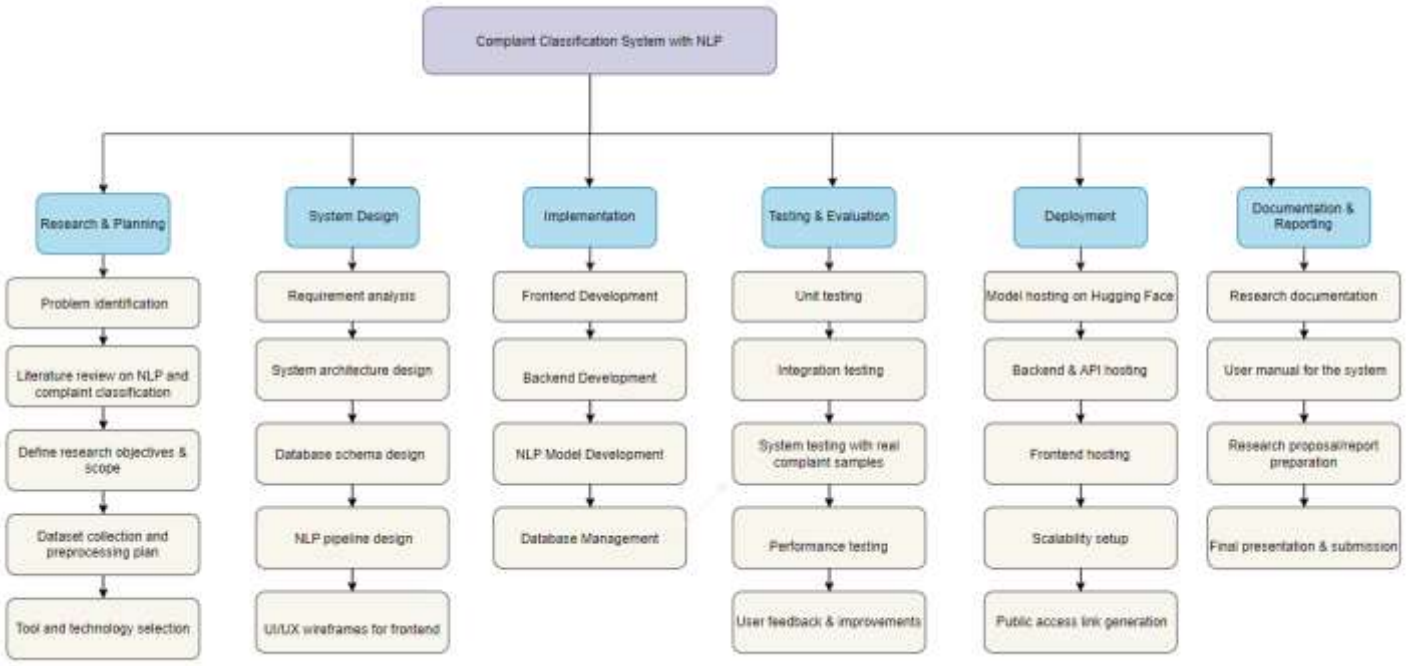

Figure 3: Work break down structure

5. GANTT CHART

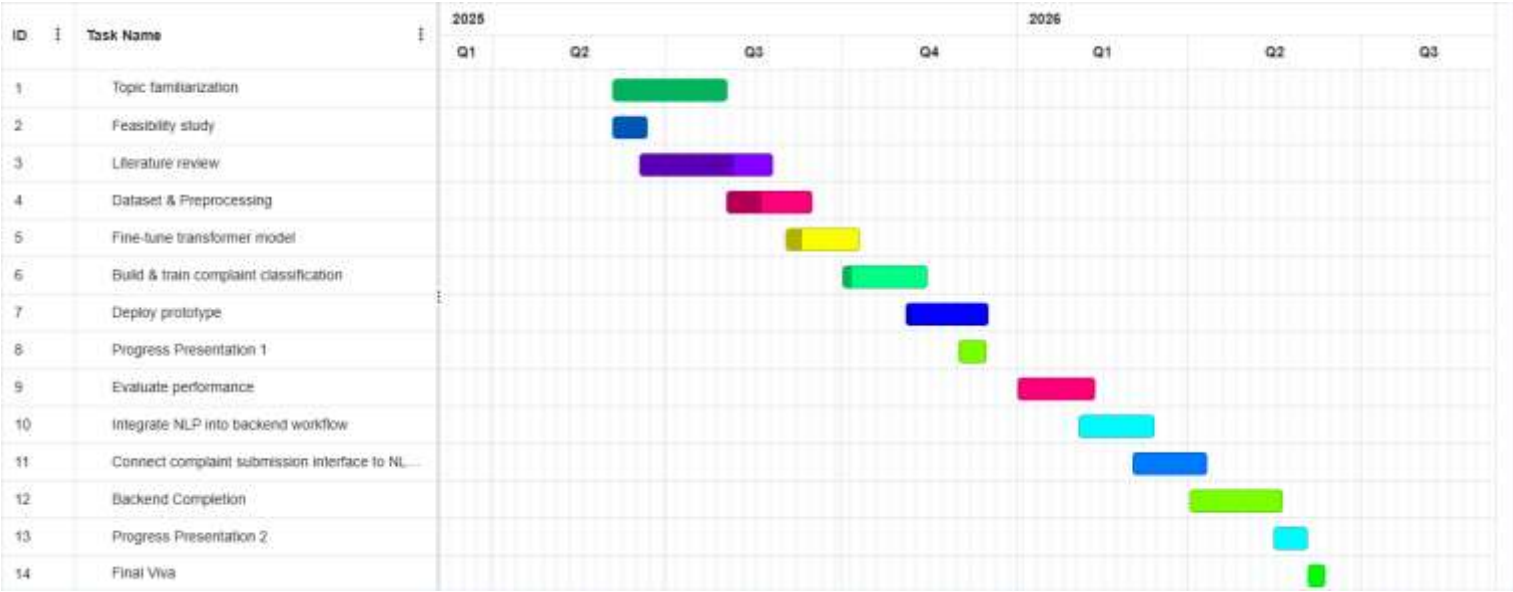

Figure 4: Gantt Chart

6. COMMERCIALIZATION OF THE PRODUCT

The government connects commercialization strategy for the Emotion Monitoring and Stage-Based Learning Optimization component emphasizes collaboration with state agencies, national education bodies, and international partners to ensure wide-scale deployment. At the core, the Ministry of Education will be engaged to integrate the solution into the national special needs and inclusive education framework, thereby reaching students with hearing impairments across Sri Lanka. This partnership ensures that the solution is not just a private-sector tool, but a government-recognized learning support system, giving it credibility and trust among educators and parents. The Ministry of Technology and ICTA will serve as enablers for digital infrastructure support, aligning the product with the broader "Digital Sri Lanka" and smart school initiatives. Additionally, the Ministry of Health will be approached to highlight the tool's secondary role in enhancing mental well-being and emotional support for hearing-impaired learners, ensuring a holistic approach to education and wellness. The rollout will begin with pilot projects in government schools for the deaf, working with SLIIT and local deaf communities to generate evidence-based outcomes on learning improvement and student engagement. Once validated, this will pave the way for Ministry-level endorsement and inclusion in teacher training programs under the National Institute of Education, ensuring teachers are fully equipped to use the system.

Funding and commercialization will be supported by multiple government-linked channels, including research and innovation grants from the National Science Foundation, ICTA's technology development programs, and international education grants from UNESCO, UNICEF, and the World Bank, accessed via government endorsements. A Public–Private Partnership (PPP) model will balance sustainability: the government can subsidize licenses for public schools to ensure inclusivity, while private schools, e-learning platforms, and special needs organizations adopt the tiered subscription model. To further enhance reach, policy alignment with the Sri Lankan Special Needs Education Policy, the UN Convention on the Rights of Persons with Disabilities (UNCRPD), and Sustainable Development Goal 4 (Quality Education) will be emphasized, ensuring that the product is seen as both a technological innovation and a policy-compliant accessibility solution. Over time, government connections will also open pathways for regional expansion into South Asian countries facing similar challenges in accessibility and sign language learning. By embedding the solution within Sri Lanka's policy framework, government infrastructure, and funding mechanisms, this commercialization strategy ensures scalability, inclusivity, profitability, and long-term sustainability while positioning Sri Lanka as a regional leader in inclusive EdTech solutions.

7. BUDGET AND BUDGET JUSTIFICATION

The proposed GovConnect: A Smart ICT Framework for Automated Civil Complaint Management and Citizen Engagement in Sri Lanka is to be implemented in a cost-effective manner. Being a university research project, it makes heavy use of open-source tools (FastAPI, TensorFlow, PyTorch, React.js, Node.js, MongoDB, etc) and only spends on essential items like dataset preparation, storage, cloud hosting and testing.

The largest share (30%) of the budget is spent on the preparation and annotation of the dataset as it is resource intensive to create a Sinhala-English code-mixed complaint dataset. This includes gathering data, cleaning it, labeling it for urgency and category ("low", "high", "abrupt", "acute"), fake resolution time for ML models, etc. The second highest allocation (20%) is to cloud computing and hosting, including domain registration, servers deployment (AWS/Render/Vercel) and GPU-based training for NLP and ML models. This guarantees scalable real-time complaint processing.

Development tools (10%) - kept small, but some money for paid plugins or APIs to be used, relying on free tools (VS Code, GitHub, Postman) Database & storage (15%) MongoDB Atlas with paid upgrades to scale to processing complaint logs, analytics, and historical data.

Testing & evaluation (10%) - encompasses usability tests, cross-browser & mobile device testing, accessibility testing and model performance testing. Internet (5%) budget for stable internet that is needed to train the model and develop for the cloud. Finally, Miscellaneous & contingency (10%) includes unexpected costs such as new APIs, libraries or additional cloud storage.
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The estimated budget of 100,000 LKR is reasonable and sustainable for a one-year student project, making it possible for GovConnect to be successfully developed, tested and deployed whilst bringing both research and practical value for public complaint management in Sri Lanka.

Category	Estimated Cost (LKR)
Dataset Preparation & Annotation	30,000
Internet Cost	5,000
Cloud Computing & Hosting	20,000
Development Tools	10,000
Database & Storage	15,000
Testing & Evaluation Tools	10,000
Miscellaneous & Contingency	10,000
Total	100,000

Table 4: Budget Plan

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